

# Model-agnostic Feature Selection via Adaptive Minipatch Ensembles

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Lili Zheng

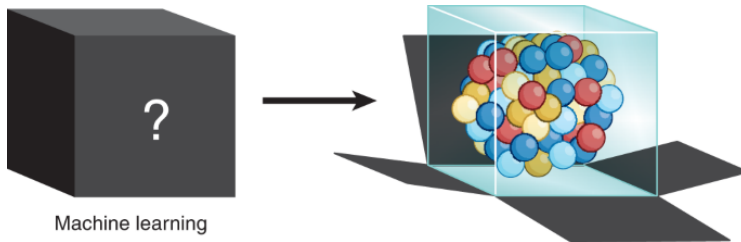
Department of Statistics

University of Illinois Urbana-Champaign

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# Extracting Interpretation from Black-box Models



- Interpretability drives scientific discovery
- E.g., which biomarkers are related to the Alzheimer's disease?

# Model-agnostic Feature Selection

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# Model-agnostic Feature Importance and Selection

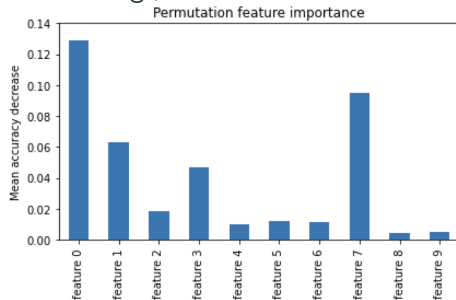
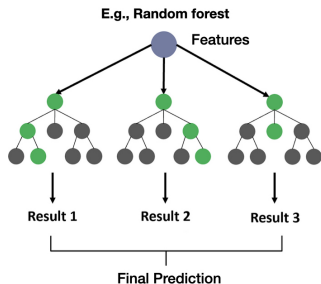
**Goal:** given any machine learning model, output important features.

- Model-X approaches (e.g., knockoff): requires knowing joint distribution of all features (Candès et al 2018)

# Model-agnostic Feature Importance and Selection

**Goal:** given any machine learning model, output important features.

- Model-X approaches (e.g., knockoff): requires knowing joint distribution of all features (Candès et al 2018)
- Recent surge of feature importance literature: e.g., leave-one-covariate-out.



- We ask: can we go from feature importance to feature selection?

# **LOCO and LOCO-MP: from Feature Importance to Feature Selection**

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**Leave-One-Covariate-Out (LOCO) feature importance** [Lei et al., 2018, Rinaldo et al., 2019]



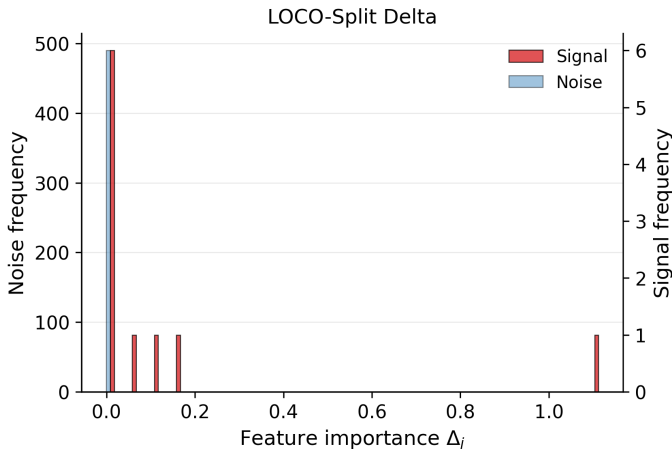
(Test error of reduced model  $\mu_{-j}$ ) - (Test error of full model  $\mu$ ).

Split data: training set  $\Rightarrow \mu, \mu_{-j}$ ; test set used to assess prediction error change.

# LOCO Feature Importance

## Demo: LOCO Importance Score (via Splitting):

Linear model, 500 features,  
200 samples, sparsity  $s = 10$ ;  
AR(1) correlation structure  
with  $\rho = 0.9$ .  
Lasso as the predictive model.





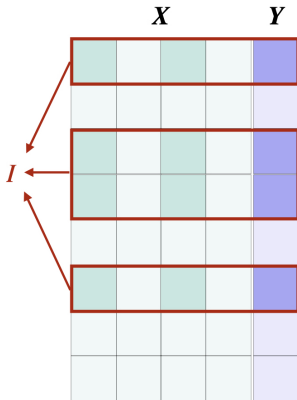
Why LOCO importance has poor separability for signal and noise features?

- Data-splitting sacrifices statistical efficiency
- Correlated features have close to zero importance

Another challenge: requires model-refitting for each feature

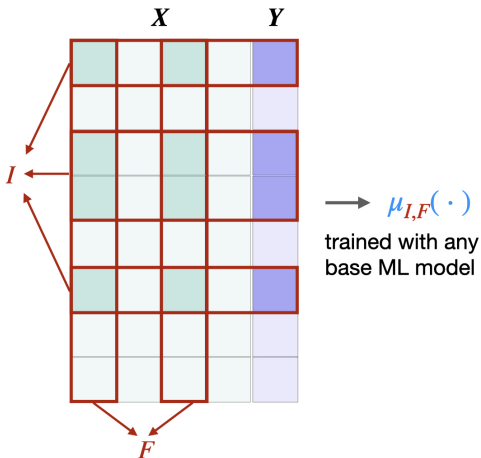
# LOCO-MP: An Ensemble Framework with Free LOCO Importance

**Minipatch ensembles:** from bagging to double-subsampling (observations and features) [Yao and Allen, 2020].



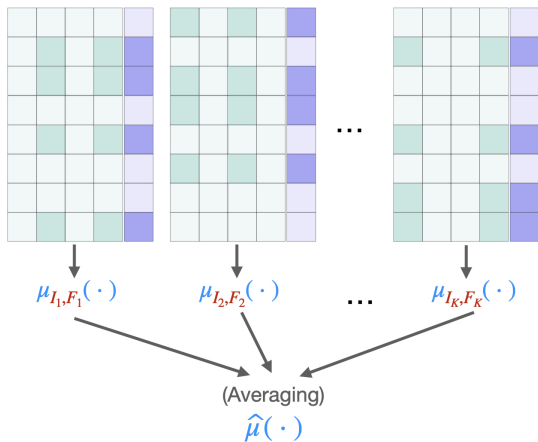
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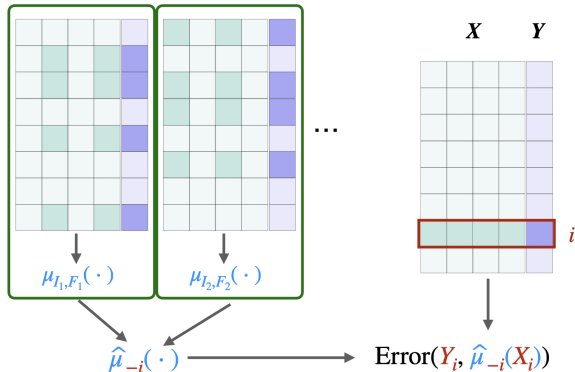


Flexible ensemble framework: can be implemented with any base model  
Memory efficient & implicit regularization & stable

# LOCO-MP: An Ensemble Framework with Free LOCO Importance

## Free LOCO Importance:

- Ensemble minipatches **without observation  $i$** .
- Compute test error on sample  $i$ .

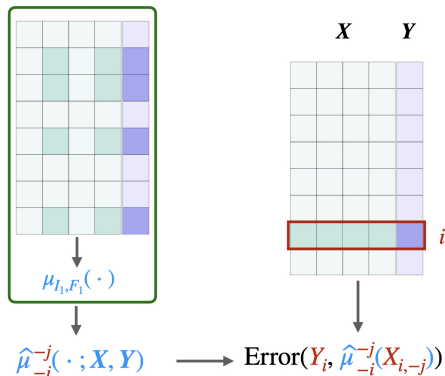


Collect and average  
**LOO**(leave-one-  
observation-out) errors

# LOCO-MP: An Ensemble Framework with Free LOCO Importance

## Free LOCO Importance:

- Ensemble minipatches **without observation  $i$  and without feature  $j$** .
- Compute test error on sample  $i$ .



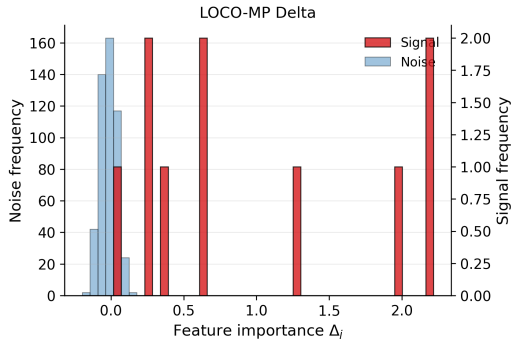
Collect and average  
**LOCO-LOO** errors

## Compared to original LOCO via Data Splitting:

- All samples used for training and evaluation
- Feature subsampling helps correlated features stand out
- Computationally free (post-training)

## Demo: LOCO Importance Score (via Minipatch Ensembling):

Same linear model setup  
Least squares as base  
model applied to each  
minipatch



Separation between signal and noise features still not clear enough;  
hard to find universal cutoff point.



## Bottlenecks for LOCO-MP:

- **Uniform** sampling of features: probability of sampling each feature is low;
- Signal is buried in noise.

**What if we iteratively update feature sampling probabilities?**

Prior adaptive sampling schemes:

- AdaSTAMP (Yao and Allen 2021): linear model feature selection
- RaSE (Tian and Feng 2021): feature screening
- Liu and Ročková (2021): feature selection formulated as a bandit, regret minimization problem

All these methods update sampling probability based on binary (selection indicator) outputs.

Not applicable for our goal: adaptive sampling given numerical feature importance scores.

We develop a new adaptive sampling scheme tailored to LOCO importance.

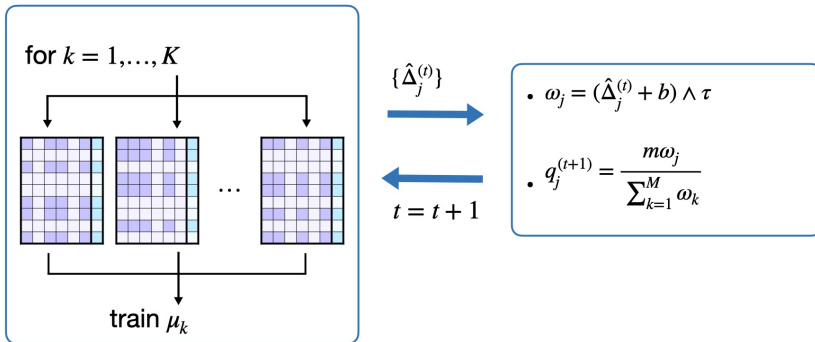
Sampling scheme:  $\mathbf{1}(j \in F)$  follows independent Bernoulli( $q_j$ ).

- $q_j$  should be larger if  $\Delta_j$  is large  $\Rightarrow q_j^{(t+1)} \propto \Delta_j^{(t)}$ .
- Minipatch size should be controlled:  $\sum_{j=1}^M q_j = m$ .
- An overly strong feature should not decrease other signal features' sampling probability  $\max_j q_j \leq \delta < 1$ .

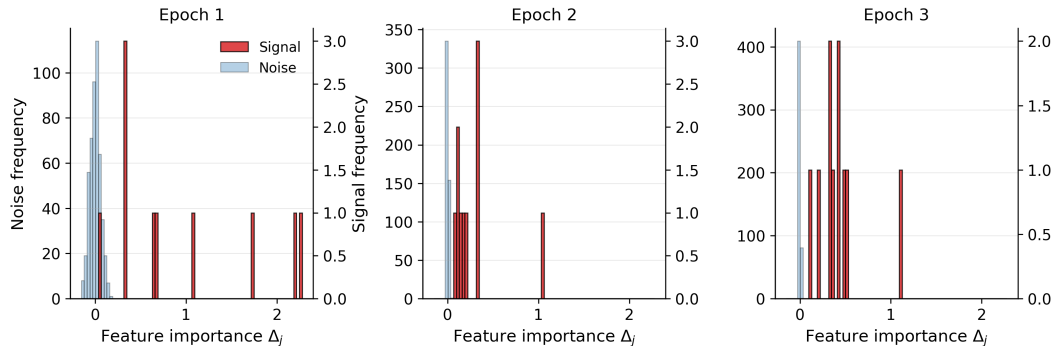
$$\Rightarrow q_j = \frac{\Delta_j \wedge \tau}{\sum_k (\Delta_k \wedge \tau)} \text{ where } \tau \text{ can be solved easily.}$$

# Adaptive Minipatch Ensembles for Feature Selection

Train minipatch ensemble with  
feature sampling prob  $q^{(t)}$

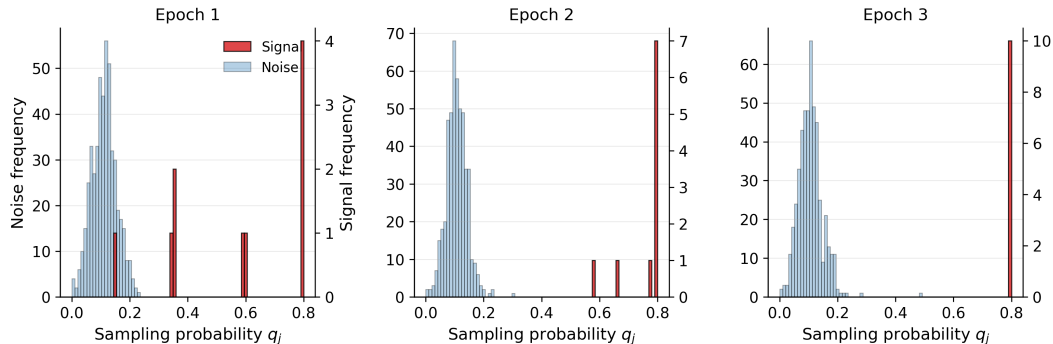


# Demo: Adaptive Sampling for Minipatch Ensembles



Stronger separation between noise and signal features!

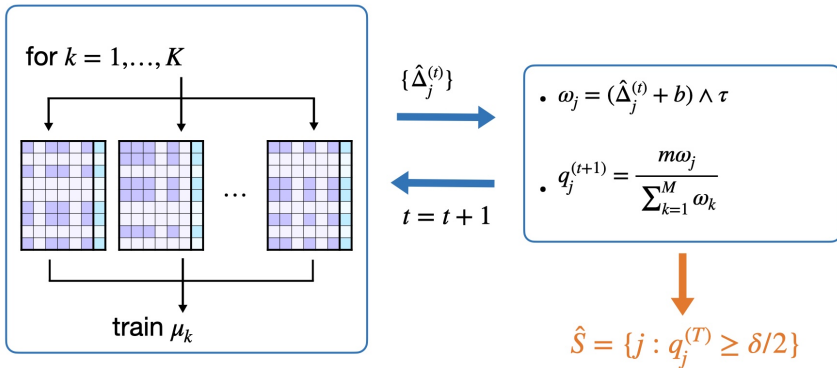
# Demo: Adaptive Sampling for Minipatch Ensembles



Also, sampling probability has a clear separation with a fixed scale.

# Demo: Adaptive Sampling for Minipatch Ensembles

Train minipatch ensemble with  
feature sampling prob  $q^{(t)}$



Here,  $\delta$  is the maximum sampling probability we set for all features.

Consider an additive model  $Y = \sum_{j \in S} f_j^*(X_j) + \varepsilon$  with independent features. Let  $f_{\min} = \min_{j \in S} \|f_j^*\|_2$

A1. Bounded minipatch predictions  $\mu_{I,F}(X)$  and bounded response  $|Y|$ .

A2.  $C(\log M \vee s) \leq m \leq cM$

A3.  $\frac{n}{N} \leq (\log M + \log N)^{-1} f_{\min}^4$

A4.  $K \geq C \frac{M}{m} (\log M + \log N) f_{\min}^{-4}$

A5. Small training error of bagged predictor  $\frac{1}{\binom{N}{n}} \mu_{I,F}(\cdot)$ , averaged over randomly sampled feature set  $F$ .



## Main Theorem (Informal)

Under Assumptions A1-A5, with high probability, the following hold:

- $q_j^{(t+1)} \geq (1 + c)q_j^{(t)} \Rightarrow S \subset \hat{S}$  if the number of iterations  $T \geq C \log(\frac{M}{m})$
- $\forall t, q_j^{(t)} \leq \delta/2 \Rightarrow \hat{S} \subset S$ .

Consequently,  $\hat{S} = S$ .

Similar results hold for general models with interactions.

How does AdaMP work for correlated features?

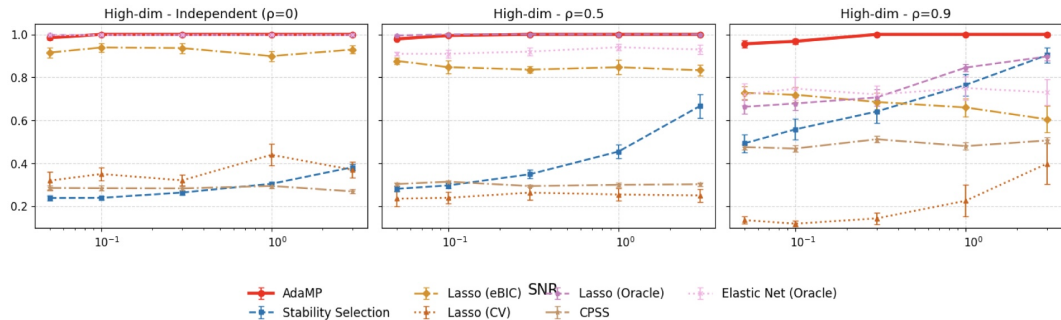
$$Y = X_1\beta + \varepsilon, \text{Cor}(X_1, X_2) = \rho, \{X_1, X_2\} \perp X_3 \perp X_4 \cdots \perp X_M.$$

### Proposition

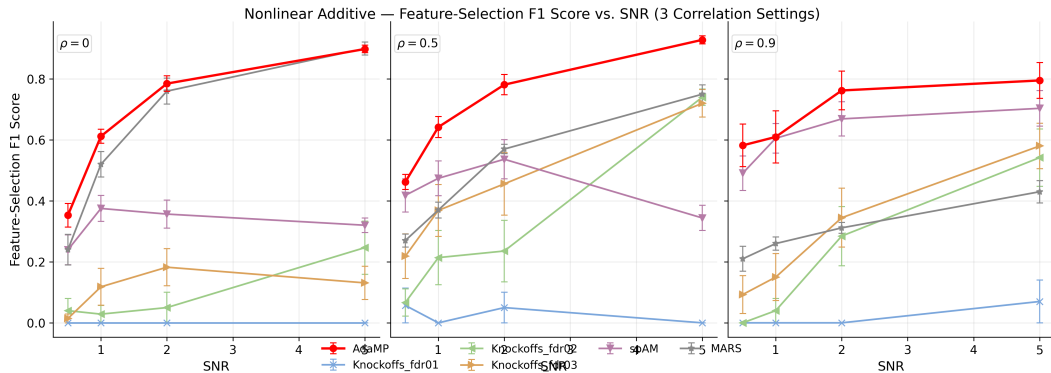
1. Even if  $\rho = \pm 1$ ,  $1 \in \hat{S}$  under mild assumptions.
2. There exists  $0 < c < 1$ , As long as  $|\rho| \leq c$ ,  $2 \notin \hat{S}$ .

- Point 1 is non-trivial, as LOCO-type metric often fails to identify correlated signal features.
- Intuition: sampling probability of any feature is bounded by  $\delta < 1$ .

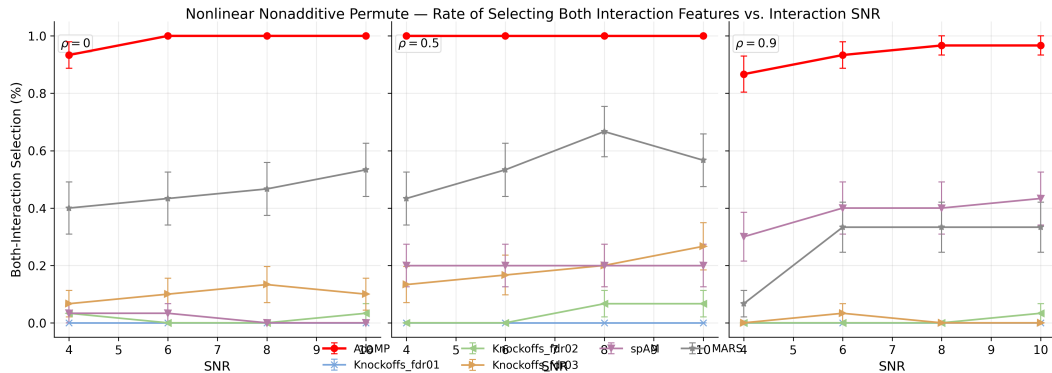
## Linear setting comparison:



## Nonlinear additive:



## Nonlinear interaction: detection rate of interaction features



- Response: riboflavin production rate (log-transformed).
- $M = 4888$  features (gene expression),  $N = 71$  observations.
- Noted by prior works: YXLD\_at and YXLE\_at are highly correlated (0.978) co-regulators.
- Base model for AdaMP: Lasso
- Compare with: Lasso, Elastic Net, CPSS, Stability Selection

## Case Studies (Riboflavin)

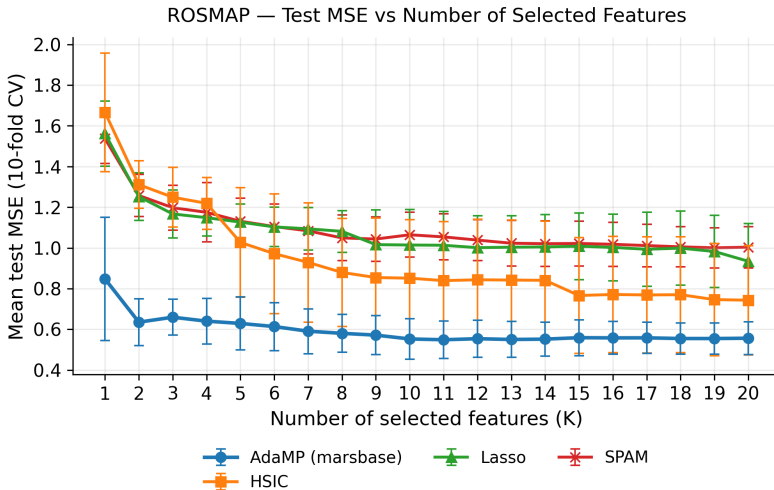
| Method                                               | $ \hat{S} $ | YOAB_at | YXLD_at | YXLE_at |
|------------------------------------------------------|-------------|---------|---------|---------|
| CPSS (Shah and Samworth, 2013)                       | 0           | —       | —       | —       |
| Elastic Net (CV) (Zou and Hastie, 2005)              | 73          | ✓       | ✓       | ✓       |
| Lasso (CV) (Tibshirani, 1996)                        | 41          | ✓       | ✓       | ✓       |
| Lasso (EBIC) (Chen and Chen, 2008)                   | 1           | —       | —       | —       |
| Stability Selection (Meinshausen and Bühlmann, 2010) | 2           | ✓       | ✓       | —       |
| AdaMP (proposed)                                     | 3           | ✓       | ✓       | ✓       |

AdaMP selects a compact set of features and does not miss the correlated pair.

- Use data from Religious Orders Study and Rush Memory and Aging Project (ROSMAP).
- Goal: predict cognition score using gene expression data.  $M = 200$  genes,  $N = 507$  samples.
- Base model for AdaMP: MARS
- Compare with: Lasso, SPAM, HSIC.



# Case Studies (ROSMAP)



Prediction error of a fitted random forest given the top- $K$  features selected by each

method

We propose a new ensemble framework for feature selection in regression which

- can be applicable to any base model outputting predictions
- does not require knowing joint feature distribution
- enjoys strong performance across linear and nonlinear settings, especially with **correlated features**.
- enjoys exact selection guarantees under mild assumptions in high dimensions.

Open questions:

- Extend feature selection to feature interaction detection
- A holistic theoretical characterization for correlated setting
- Incorporate minipatch framework with large ML models: apply double subsampling only during fine-tuning?

**Thank you!**

## References

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- Alessandro Rinaldo, Larry Wasserman, and Max G'Sell. Bootstrapping and sample splitting for high-dimensional, assumption-lean inference. *The Annals of Statistics*, 47(6):3438–3469, 2019.
- Tianyi Yao and Genevera I Allen. Feature selection for huge data via minipatch learning. *arXiv preprint arXiv:2010.08529*, 2020.